

Drift-Aware Selective Prediction for Reliable Transformer Classification Under Distribution Shift

Corresponding Author: Shivaansh Sharma

Student at Department of Computer Science and Engineering,
Punjab Engineering College (Deemed to be University),
Chandigarh, India

Email: shivaanshsharma.business@gmail.com

Orcid ID: 0009-0001-4977-7385

Abstract

Transformer-based text classification models often exhibit overconfident predictions under semantic distribution shift, limiting their reliability in safety-critical and real-world deployment settings. Although post-hoc calibration techniques improve probability estimation, they typically force predictions even when input samples deviate substantially from the training distribution. This work introduces a lightweight inference-time framework, termed Drift-Aware Selective Calibration (DASC), for reliability-aware selective prediction under semantic drift. The proposed approach combines embedding-space drift estimation using Mahalanobis distance with adaptive confidence modulation to selectively abstain from high-risk predictions. Experiments conducted using DistilBERT trained on the AG News dataset and evaluated under both in-domain and out-of-domain settings demonstrate that semantic drift increases significantly under distribution shift, while predictive confidence remains persistently overconfident. In out-of-domain evaluation using Yahoo Answers, the model maintained high average confidence (0.91) despite substantial drift escalation, revealing confidence unreliability under semantic shift. DASC successfully reduced operational trust under drift by lowering coverage from 95.0% in-domain to 67.5% under out-of-domain conditions while improving selective prediction accuracy up to 96.7% for high-confidence retained samples. Sensitivity analysis further reveals a controllable reliability–coverage tradeoff governed by drift sensitivity. The findings demonstrate that integrating semantic drift estimation with selective prediction provides an effective and computationally lightweight mechanism for improving transformer reliability under distribution shift without requiring retraining or architectural modification.

Keywords

Transformer reliability; Selective prediction; Semantic drift; Distribution shift; Confidence calibration; Drift-aware abstention; Mahalanobis distance; Out-of-distribution detection; Trustworthy AI; Transformer classification; Risk-coverage analysis; Uncertainty estimation

1 Introduction

Transformer-based architectures have become the dominant paradigm for natural language processing tasks, including text classification, question answering, summarization, and code generation. Models such as BERT, DistilBERT, and their variants achieve high predictive accuracy through large-scale pretraining and contextual representation learning. Despite these advances, modern transformer models often exhibit poor uncertainty estimation and overconfident behaviour, particularly under distribution shift. In real-world deployment scenarios, input data frequently deviates from the training distribution due to domain variation, semantic drift, evolving language patterns, or unseen contextual structures. Under such conditions, transformer models may produce highly confident yet unreliable predictions, posing risks in safety-critical and decision-support applications.

Confidence calibration has emerged as a central research direction for improving probabilistic reliability in deep neural networks. Post-hoc calibration methods such as temperature scaling attempt to align predictive confidence with empirical correctness likelihood without retraining the underlying model. Although these methods improve in-domain calibration, they typically apply global confidence correction and still force prediction for all inputs, including semantically shifted samples. Consequently, calibrated confidence may remain operationally unreliable under substantial distributional deviation.

Recent studies in trustworthy artificial intelligence have increasingly emphasized selective prediction, where models are permitted to abstain from uncertain predictions. Selective prediction frameworks attempt to improve reliability by rejecting low-confidence samples. However, most existing approaches rely primarily on softmax confidence or entropy-based uncertainty estimation. This assumption becomes problematic under semantic distribution shift, where transformer models often remain overconfident despite substantial deviation from the training distribution. As a result, confidence-only abstention mechanisms may fail to detect semantically unreliable predictions.

Representation-space geometry provides an alternative perspective for understanding distribution shift. Transformer embedding spaces encode semantic structure, and deviations from the training distribution can be quantified through distance-based measures. In particular, Mahalanobis distance offers a covariance-aware measure of embedding-space drift and has been widely used for out-of-distribution detection. Nevertheless, prior work primarily employs such methods for binary anomaly detection rather than integrating semantic drift directly into selective prediction and reliability-aware abstention.

This work investigates the following research question:

How can transformer models selectively abstain from unreliable predictions under semantic distribution shift when confidence estimates remain overconfident?

To address this problem, we propose Drift-Aware Selective Calibration (DASC), a lightweight inference-time framework for reliability-aware selective prediction under semantic drift. The proposed approach combines Mahalanobis embedding-space drift estimation with adaptive confidence modulation to selectively reduce operational trust for semantically shifted samples. Unlike conventional confidence thresholding, the proposed

framework conditions selective prediction on both predictive confidence and representation-space drift, enabling drift-aware abstention without retraining or architectural modification.

Experiments are conducted using DistilBERT trained on the AG News dataset and evaluated under both in-domain and out-of-domain settings using Yahoo Answers as a shifted-domain benchmark. Empirical analysis demonstrates that semantic drift increases substantially under distribution shift, while predictive confidence remains persistently overconfident. The proposed DASC framework selectively abstains from high-drift predictions, improving selective reliability while revealing a controllable reliability–coverage tradeoff governed by drift sensitivity.

The main contributions of this work are summarized as follows:

1. We introduce Drift-Aware Selective Calibration (DASC), a lightweight selective prediction framework that integrates semantic drift estimation with adaptive confidence modulation for transformer reliability under distribution shift.
2. We demonstrate empirically that transformer confidence remains strongly overconfident under semantic drift, even when representation-space deviation increases substantially.
3. We show that combining Mahalanobis drift estimation with selective prediction enables operational reliability control through drift-aware abstention.
4. We provide risk–coverage and sensitivity analyses demonstrating a controllable tradeoff between selective accuracy, abstention behaviour, and reliability under varying drift sensitivity levels.
5. We evaluate the proposed framework under both in-domain and out-of-domain conditions, demonstrating effective selective abstention under semantic distribution shift without requiring retraining or additional model parameters.

Unlike prior confidence-based selective prediction approaches, the proposed DASC framework integrates representation-space semantic drift directly into confidence modulation for transformer abstention under distribution shift. In contrast to retraining-based uncertainty methods, DASC operates entirely at inference time without requiring additional supervision, ensemble architectures, or calibration retraining. The framework therefore provides a lightweight and operationally practical mechanism for reliability-aware transformer prediction under semantic drift.

2 Related Work

2.1 Confidence Calibration in Deep Neural Networks

Confidence calibration aims to align predictive probabilities with empirical correctness likelihood. Modern deep neural networks, including transformer architectures, are frequently overconfident despite achieving high classification accuracy. Guo et al. demonstrated that contemporary neural networks are often poorly calibrated and introduced temperature scaling as a simple yet effective post-hoc calibration method. Since then, several extensions have been proposed, including vector scaling, matrix scaling, Dirichlet calibration, focal-loss-based calibration, Bayesian uncertainty estimation, and ensemble-based approaches.

Although post-hoc calibration methods improve probabilistic alignment under in-domain evaluation, they typically assume that validation and deployment distributions are similar. Under semantic or covariate shift, confidence estimates may degrade substantially. Moreover, most calibration methods still force predictions for all inputs, even when uncertainty becomes unreliable under distribution shift. Consequently, calibrated confidence alone may not provide sufficient operational reliability in real-world deployment settings.

2.2 Selective Prediction and Abstention

Selective prediction, also referred to as classification with abstention or reject-option learning, allows models to abstain from uncertain predictions in order to improve reliability. Early work in selective classification introduced confidence thresholding mechanisms that reject low-confidence samples. More recent approaches, including SelectiveNet and risk-coverage optimization frameworks, explicitly optimize the tradeoff between predictive coverage and selective risk.

Most existing selective prediction methods rely primarily on softmax confidence, entropy, or uncertainty estimates derived from model probabilities. However, transformer models often remain highly confident even when predictions are semantically unreliable under distribution shift. This limitation becomes particularly problematic in out-of-domain settings where confidence may fail to reflect representation-level deviation from the training distribution.

Our work differs from prior selective prediction approaches by explicitly conditioning abstention on representation-space semantic drift rather than confidence alone. Instead of treating uncertainty solely as a probability estimation problem, we integrate embedding-space distributional deviation directly into the selective prediction process.

2.3 Distribution Shift and Transformer Reliability

Distribution shift remains one of the central challenges in trustworthy machine learning. Prior research has shown that transformer models exhibit degraded calibration, increased prediction instability, and overconfidence under domain shift, covariate shift, and semantic drift. Ovadia et al. demonstrated that uncertainty estimation methods deteriorate significantly under real-world dataset shift, highlighting limitations of conventional confidence estimation approaches.

Several approaches have attempted to improve robustness under shift through domain adaptation, uncertainty regularization, adversarial training, or retraining-based calibration strategies. However, these methods often require access to shifted-domain data during optimization or introduce substantial computational overhead.

In contrast, our framework operates entirely at inference time without retraining or architectural modification. The proposed approach focuses specifically on operational reliability by selectively reducing trust in semantically drifted samples.

2.4 Representation-Space Drift and Mahalanobis Distance

Representation-space geometry provides an effective mechanism for analyzing semantic deviation in deep neural networks. Distance-based methods have been widely applied for out-of-distribution detection, anomaly detection, and uncertainty estimation. Among these approaches, Mahalanobis distance has emerged as a particularly effective covariance-aware measure of representation-space deviation. Lee et al. demonstrated that Mahalanobis distance can successfully identify out-of-distribution samples using intermediate neural representations.

Subsequent studies extended representation-based drift estimation to transformer architectures and embedding-space analysis. However, most prior work employs Mahalanobis distance primarily for binary out-of-distribution detection or anomaly scoring. The integration of representation-space drift directly into selective prediction and abstention mechanisms remains comparatively underexplored.

This work extends representation-based drift analysis beyond anomaly detection by using semantic drift as a continuous reliability modulation signal for selective prediction.

2.5 Research Gap

Prior work has extensively studied confidence calibration, uncertainty estimation, out-of-distribution detection, and selective prediction independently. Nevertheless, limited research has unified these perspectives into a lightweight inference-time reliability framework for transformer models under semantic distribution shift.

In particular, existing selective prediction methods largely rely on confidence estimates without explicitly accounting for representation-space semantic deviation. At the same time, representation-based drift estimation approaches are typically used only for anomaly detection rather than operational abstention and reliability-aware prediction control.

To address this gap, we propose Drift-Aware Selective Calibration (DASC), which integrates Mahalanobis embedding-space drift estimation with adaptive confidence modulation for selective prediction under semantic drift. The proposed framework enables transformer models to selectively abstain from semantically unreliable predictions while maintaining lightweight inference-time deployment.

3 Methodology

3.1 Problem Formulation

Let a transformer-based classifier produce logits:

$$z(x) \in \mathbb{R}^K$$

for an input sample x , where K denotes the number of output classes. Predictive probabilities are computed using the softmax function:

$$p(y | x) = \text{softmax}(z(x)).$$

Traditional classification systems always produce a prediction regardless of uncertainty or distributional deviation. However, under semantic distribution shift, transformer models may generate highly confident yet unreliable predictions. The objective of this work is therefore not only to improve confidence estimation, but also to determine when the model should abstain from prediction.

We formulate this as a selective prediction problem under semantic drift, where the model is permitted to reject uncertain or semantically shifted samples.

3.2 Transformer Classification Framework

We employ DistilBERT as the backbone transformer encoder for text classification. Given an input sequence:

$$x = (w_1, w_2, \dots, w_n),$$

the transformer produces contextual hidden representations:

$$H = (h_1, h_2, \dots, h_n).$$

The contextual embedding corresponding to the CLS token is extracted as the global semantic representation:

$$f(x) = h_{\text{CLS}} \in \mathbb{R}^d,$$

where d denotes embedding dimensionality.

The classification head maps the CLS embedding to output logits:

$$z(x) = Wf(x) + b.$$

Predictive confidence is defined as:

$$c(x) = \max_k p(y = k \mid x).$$

3.3 Semantic Drift Estimation

To quantify semantic deviation from the training distribution, we compute embedding-space drift using Mahalanobis distance.

Let the training embedding distribution be characterized by mean vector:

$$\mu = \mathbb{E}[f(x)]$$

and covariance matrix:

$$\Sigma = \text{Cov}(f(x)).$$

To ensure numerical stability during inversion, covariance regularization is applied:

$$\Sigma_\epsilon = \Sigma + \epsilon I,$$

where $\epsilon > 0$ is a small constant and I is the identity matrix.

For a test sample x , semantic drift is defined as the Mahalanobis distance:

$$D(x) = \sqrt{(f(x) - \mu)^T \Sigma_\epsilon^{-1} (f(x) - \mu)}.$$

Mahalanobis distance is preferred over Euclidean distance because it accounts for covariance structure and correlated embedding dimensions.

Drift scores are normalized to the interval $[0,1]$:

$$\tilde{D}(x) = \frac{D(x) - D_{\min}}{D_{\max} - D_{\min}}.$$

Normalization enables stable drift-aware confidence modulation across datasets and experimental conditions.

3.4 Drift-Aware Selective Calibration (DASC)

The central hypothesis of this work is that predictive confidence should decrease as semantic drift increases. Instead of relying solely on softmax confidence, we introduce Drift-Aware Selective Calibration (DASC), which integrates confidence estimation with representation-space drift.

A drift-aware robustness score is defined as:

$$R(x) = \exp(-\alpha \tilde{D}(x)),$$

where:

- $\tilde{D}(x)$ denotes normalized semantic drift,
- $\alpha \geq 0$ controls drift sensitivity.

The proposed drift-aware confidence score is then computed as:

$$S(x) = c(x) \cdot R(x).$$

Substituting the robustness term yields:

$$S(x) = c(x) \exp(-\alpha \tilde{D}(x)).$$

As semantic drift increases, the robustness term decreases exponentially, reducing operational trust even when softmax confidence remains high.

The computational overhead of DASC is minimal because Mahalanobis drift estimation operates only on fixed-dimensional CLS embeddings during inference.

3.5 Selective Prediction Mechanism

Selective prediction is performed using a threshold-based abstention mechanism.

For a predefined threshold τ :

$$\hat{y}(x) = \begin{cases} \arg \max_k p(y = k | x), & S(x) \geq \tau \\ \text{abstain}, & S(x) < \tau \end{cases}$$

where:

- $S(x)$ is the drift-aware confidence score,
- τ controls selective coverage.

Unlike conventional confidence thresholding, the proposed framework conditions prediction reliability on both confidence and semantic drift.

3.6 Drift-Risk Relationship

The proposed framework is motivated by the observation that predictive instability generally increases under semantic deviation from the training distribution.

Let selective risk denote prediction error over non-abstained samples:

$$R_{\text{sel}} = \mathbb{P}(\hat{y} \neq y \mid S(x) \geq \tau).$$

As representation-space drift increases, the likelihood of unreliable prediction also increases. Consequently, semantic drift acts as a proxy for predictive instability under distribution shift.

This motivates the use of drift-aware confidence modulation for operational reliability control.

3.7 Risk–Coverage Evaluation

To evaluate selective prediction behaviour, we analyze the relationship between predictive coverage and selective risk.

Coverage is defined as the fraction of retained predictions:

$$\text{Coverage} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(S(x_i) \geq \tau).$$

Selective accuracy is computed only over retained predictions:

$$\text{Selective Accuracy} = \frac{\sum_{i=1}^N \mathbf{1}(\hat{y}_i = y_i) \mathbf{1}(S(x_i) \geq \tau)}{\sum_{i=1}^N \mathbf{1}(S(x_i) \geq \tau)}.$$

Selective risk is defined as:

$$\text{Selective Risk} = 1 - \text{Selective Accuracy}.$$

Risk–coverage curves are constructed by varying the retention threshold τ . We further compute the Area Under the Risk–Coverage Curve (AURC), where lower values indicate better selective prediction efficiency.

3.8 Sensitivity Analysis

The drift sensitivity parameter α controls the strength of semantic drift penalization.

- Small α values produce weak drift modulation and retain higher coverage.
- Large α values aggressively penalize semantically shifted samples, increasing abstention behaviour.

Sensitivity analysis is therefore conducted across multiple α values to evaluate the reliability–coverage tradeoff induced by drift-aware selective prediction.

4 Experimental Setup

4.1 Datasets

Experiments were conducted using both in-domain and out-of-domain text classification datasets to evaluate transformer reliability under semantic distribution shift.

AG News Dataset

The AG News corpus was used as the primary in-domain text classification benchmark. The dataset consists of four news categories:

- World,
- Sports,
- Business,
- Science/Technology.

Following prior transformer-based text classification studies, a subset of the dataset was used for computational efficiency and reproducibility. Specifically:

- 12,000 samples were used for training,
- 3,000 samples were used for in-domain evaluation.

The AG News dataset provides relatively clean semantic structure and balanced class distribution, making it suitable for calibration and selective prediction analysis.

Yahoo Answers Dataset

To evaluate semantic distribution shift, the Yahoo Answers Topics dataset was used as an out-of-domain (OOD) benchmark. Unlike AG News, Yahoo Answers contains:

- conversational language,
- question–answer structures,
- broader semantic variability,
- and informal linguistic patterns.

This creates substantial semantic deviation from the news-domain training distribution and provides an effective benchmark for evaluating transformer reliability under distribution shift.

A subset of 3,000 test samples was used for OOD evaluation.

4.2 Transformer Backbone

DistilBERT was employed as the transformer encoder for all experiments due to its computational efficiency and strong text classification performance.

The pretrained model:

distilbert-base-uncased

was fine-tuned for four-class text classification using the AG News dataset.

The CLS token embedding extracted from the final hidden layer was used as the semantic representation for:

- classification,
- drift estimation,
- and selective prediction analysis.

4.3 Training Configuration

Model training was performed using the HuggingFace Transformers framework.

The hyperparameters used during fine-tuning are mentioned in Table 1

Parameter	Value
Optimizer	AdamW
Learning Rate	5×10^{-5}
Batch Size	16
Number of Epochs	2
Maximum Sequence Length	128
Weight Decay	0.01

Table 1. Fine-tuning hyperparameters used for DistilBERT training on AG News.

Training was conducted using NVIDIA GPU acceleration through CUDA-enabled PyTorch execution.

The model achieved:

- 92.4% in-domain classification accuracy,
- and 0.924 macro-F1 score
on the AG News test set.

4.4 Semantic Drift Estimation

Semantic drift was computed using Mahalanobis distance over CLS embedding representations.

Training embeddings were used to estimate:

- the mean representation vector,
- and covariance structure of the in-domain semantic space.

Covariance regularization was applied to ensure stable inversion during Mahalanobis distance computation.

Normalized drift scores were subsequently used for:

- drift analysis,
- confidence modulation,
- and selective prediction.

4.5 Drift-Aware Selective Calibration (DASC)

The proposed DASC framework combines predictive confidence with semantic drift estimation using:

$$S(x) = c(x) \exp(-\alpha \tilde{D}(x)),$$

where:

- $c(x)$ denotes softmax confidence,
- $\tilde{D}(x)$ represents normalized semantic drift,
- and α controls drift sensitivity.

Selective prediction was performed using threshold-based abstention:

$$S(x) \geq \tau.$$

Unless otherwise stated, the selective threshold was fixed at:

$$\tau = 0.70.$$

Sensitivity analysis was conducted using:

$$\alpha \in \{0.1, 0.3, 0.5, 0.7, 1.0, 1.5, 2.0\}.$$

4.6 Baseline Method

To evaluate the effectiveness of drift-aware selective prediction, the proposed framework was compared against conventional confidence thresholding.

The baseline method performs selective prediction solely using softmax confidence:

$$c(x) \geq \tau.$$

Unlike DASC, the baseline does not account for representation-space semantic drift.

This comparison enables analysis of whether semantic drift provides additional reliability information beyond predictive confidence alone.

4.7 Evaluation Metrics

Multiple evaluation metrics were used to analyze calibration quality, selective prediction behaviour, and reliability under semantic shift.

Classification Metrics

- Accuracy
- Macro-F1 Score

Reliability Metrics

- Predictive Confidence
- Selective Accuracy
- Selective Risk
- Coverage

Selective Prediction Metrics

- Risk–Coverage Curves
- Area Under the Risk–Coverage Curve (AURC)

Lower AURC values indicate more efficient selective prediction behaviour.

4.8 Experimental Objectives

The experiments were designed to investigate the following hypotheses:

1. Transformer confidence remains overconfident under semantic distribution shift.
2. Representation-space drift increases significantly under out-of-domain conditions.
3. Confidence-only selective prediction is insufficient under semantic drift.
4. Drift-aware selective prediction improves operational reliability through adaptive abstention.
5. Drift sensitivity induces a controllable reliability–coverage tradeoff.

These hypotheses collectively evaluate the effectiveness of DASC as a lightweight inference-time reliability framework for transformer models under distribution shift.

5 Results

5.1 In-Domain Classification Performance

The DistilBERT classifier achieved strong in-domain performance on the AG News dataset after fine-tuning for two epochs. The model obtained:

- 92.4% classification accuracy,

- and a macro-F1 score of 0.924.

These results indicate that the transformer model successfully learned discriminative semantic representations for the news-domain classification task. Furthermore, the high predictive confidence observed during evaluation demonstrates that the model exhibits strong certainty on in-domain samples.

The average predictive confidence on the AG News test set was:

$$\bar{c}_{ID} = 0.976.$$

Although high confidence is expected for in-domain data, such behaviour becomes problematic when models encounter semantically shifted inputs.

5.2 Semantic Drift Analysis

To analyze semantic deviation from the training distribution, Mahalanobis distance was computed over CLS embedding representations extracted from the transformer encoder.

Figure 1 illustrates the normalized drift distribution for in-domain AG News samples. Most samples remain concentrated within a low-drift region, indicating strong semantic consistency with the training distribution. However, a noticeable long-tail drift distribution is also observed, suggesting that certain samples deviate substantially from the learned embedding manifold even within the same dataset.

The mean Mahalanobis drift score for in-domain samples was:

$$\bar{D}_{ID} = 27.68.$$

These observations demonstrate that representation-space geometry captures meaningful semantic variability in transformer embeddings.

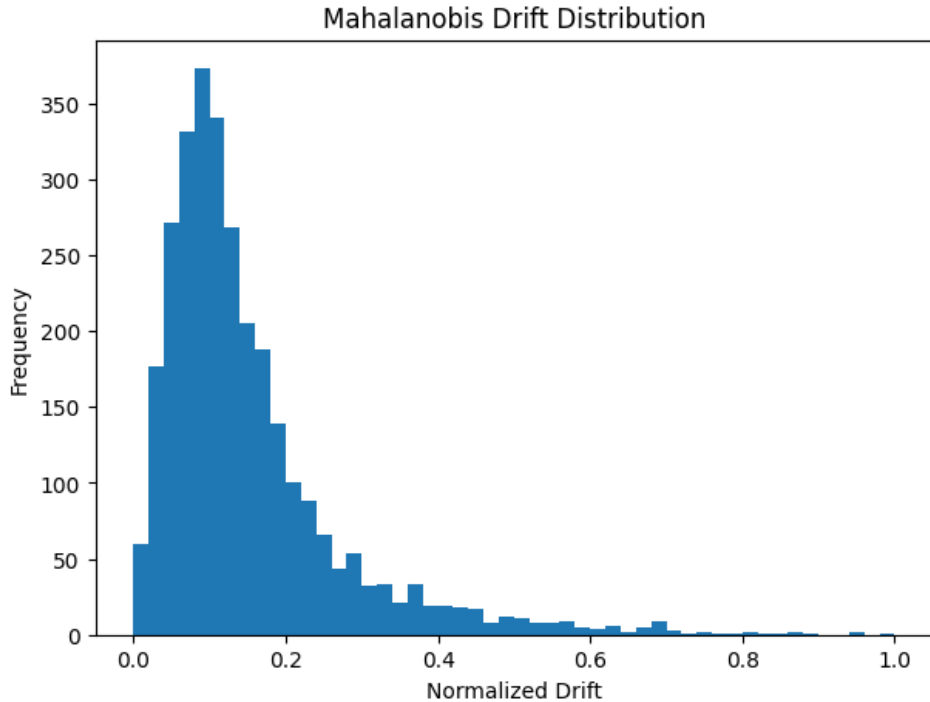


Figure 1. Normalized Mahalanobis drift distribution for in-domain AG News samples.

5.3 Confidence Under Semantic Drift

To investigate the relationship between predictive confidence and semantic drift, confidence values were plotted against normalized Mahalanobis drift scores.

Figure 2 shows that predictive confidence decreases only moderately as semantic drift increases. The computed correlation between drift and confidence was:

$$r = -0.378.$$

Although a negative correlation exists, the relationship remains relatively weak. Importantly, many high-drift samples still exhibit extremely high confidence values near 1.0.

This observation reveals a critical reliability limitation in transformer models: even under substantial semantic deviation, softmax confidence frequently remains overconfident.

Such behaviour suggests that confidence-only selective prediction mechanisms may fail to identify semantically unreliable predictions under distribution shift.

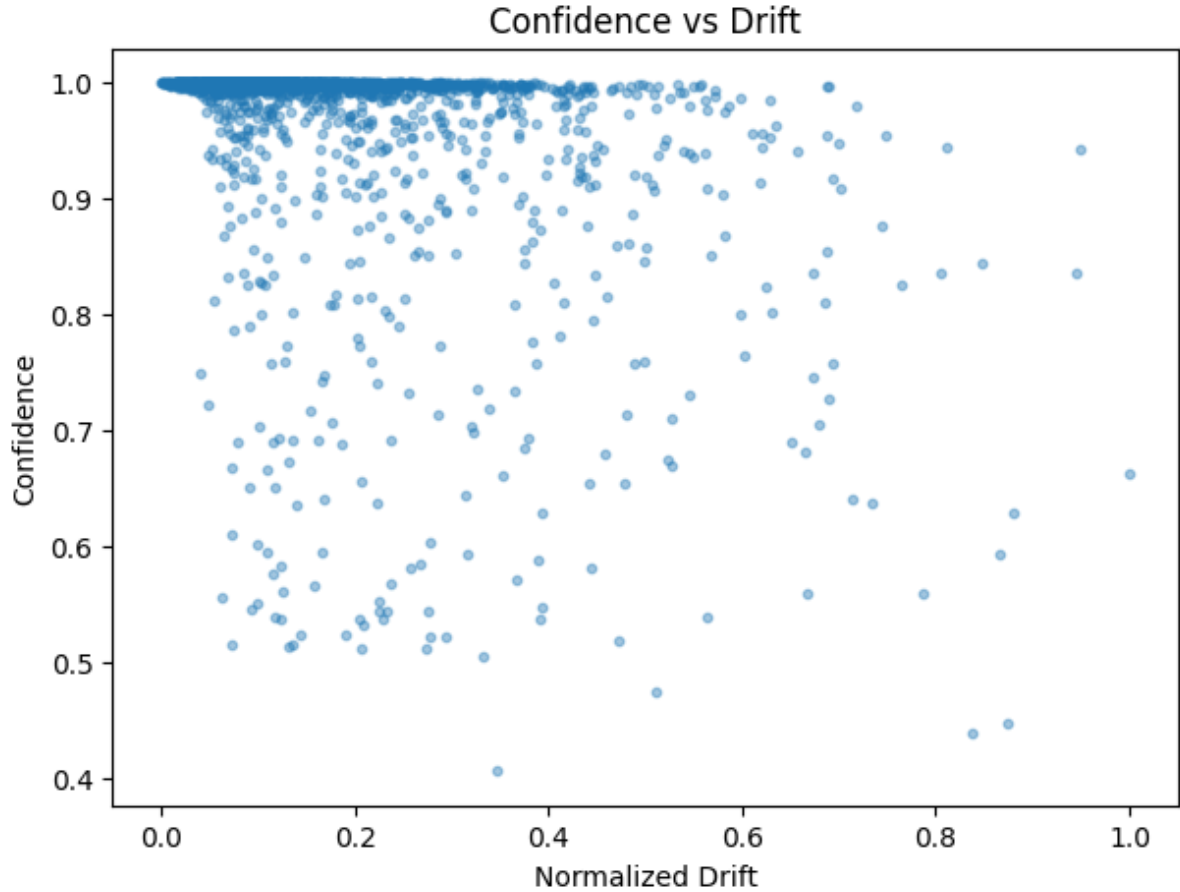


Figure 2. Relationship between predictive confidence and normalized semantic drift.

5.4 Selective Prediction Performance

We next evaluated selective prediction performance using both conventional confidence thresholding and the proposed Drift-Aware Selective Calibration (DASC) framework.

Using confidence thresholding alone with:

$$\tau = 0.70,$$

the baseline method achieved:

- 93.3% coverage,
- and 94.9% selective accuracy.

The proposed DASC framework integrates semantic drift into confidence estimation using:

$$S(x) = c(x)\exp(-\alpha\tilde{D}(x)).$$

As drift sensitivity increased, selective accuracy improved while predictive coverage decreased, indicating progressively stronger abstention behaviour.

For example, with:

$$\alpha = 2.0,$$

DASC achieved:

- 96.7% selective accuracy,
- but reduced coverage to 70.9%.

This demonstrates that drift-aware selective prediction successfully filters semantically unreliable predictions. However, aggressive drift penalization also induces conservative abstention behaviour.

Table 2 summarizes the selective prediction performance across multiple drift sensitivity levels.

Method	Coverage	Selective Accuracy	AURC
Confidence Thresholding	0.9333	0.9493	0.0141
DASC ($\alpha = 0.1$)	0.9690	0.9367	0.0164
DASC ($\alpha = 0.5$)	0.9503	0.9428	0.0204
DASC ($\alpha = 1.0$)	0.8887	0.9554	0.0229
DASC ($\alpha = 2.0$)	0.7097	0.9671	0.0252

Table 2. Selective Prediction Performance Under Varying Drift Sensitivity

The results reveal a controllable reliability–coverage tradeoff governed by drift sensitivity.

These results indicate that DASC prioritizes retained-sample trustworthiness under semantic drift rather than global risk minimization alone.

5.5 Risk–Coverage Analysis

Figure 3 presents the risk–coverage curves for both confidence thresholding and DASC.

As predictive coverage decreases, selective risk consistently declines, indicating that abstention successfully removes unreliable predictions. The proposed DASC framework demonstrates more aggressive risk reduction under lower coverage regimes due to drift-aware abstention.

However, excessive drift penalization also increases abstention frequency, leading to higher Area Under the Risk–Coverage Curve (AURC) values for large α .

The best AURC was obtained using low drift sensitivity:

$$\alpha = 0.1.$$

This finding suggests that moderate semantic drift penalization provides the most balanced operational reliability behaviour.

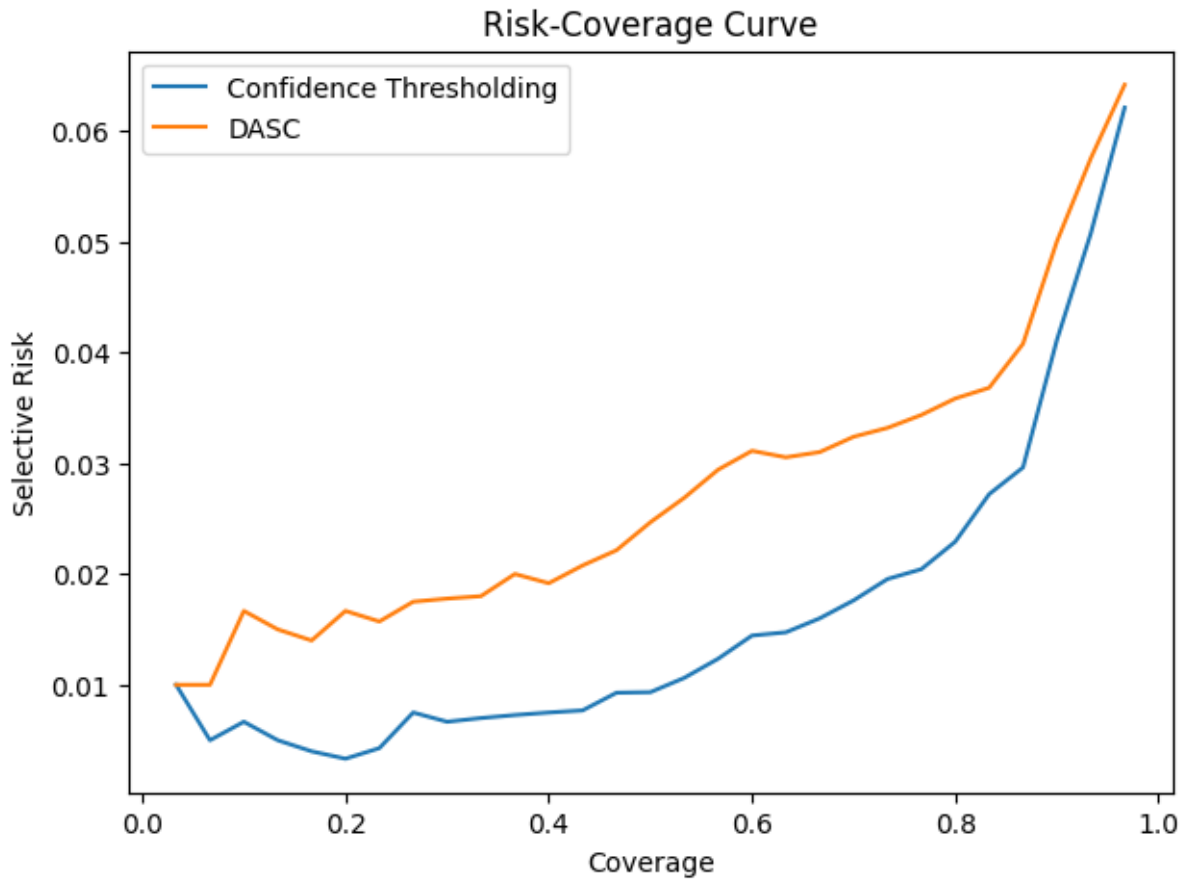


Figure 3. Risk–coverage comparison between confidence thresholding and Drift-Aware Selective Calibration (DASC).

5.6 Sensitivity Analysis

To evaluate robustness under varying drift sensitivity, experiments were conducted across:

$$\alpha \in \{0.1, 0.3, 0.5, 0.7, 1.0, 1.5, 2.0\}.$$

Figures 4–6 illustrate the effect of drift sensitivity on:

- selective accuracy,
- coverage,
- and AURC.

The results show that:

- larger α values improve selective accuracy,
- while simultaneously reducing coverage.

This behaviour confirms that drift sensitivity directly controls operational conservativeness in selective prediction.

Importantly, excessively large α values induce over-conservative abstention, reducing selective efficiency despite improved retained-sample reliability.

These findings demonstrate that DASC provides controllable reliability modulation rather than uniformly aggressive rejection behaviour.

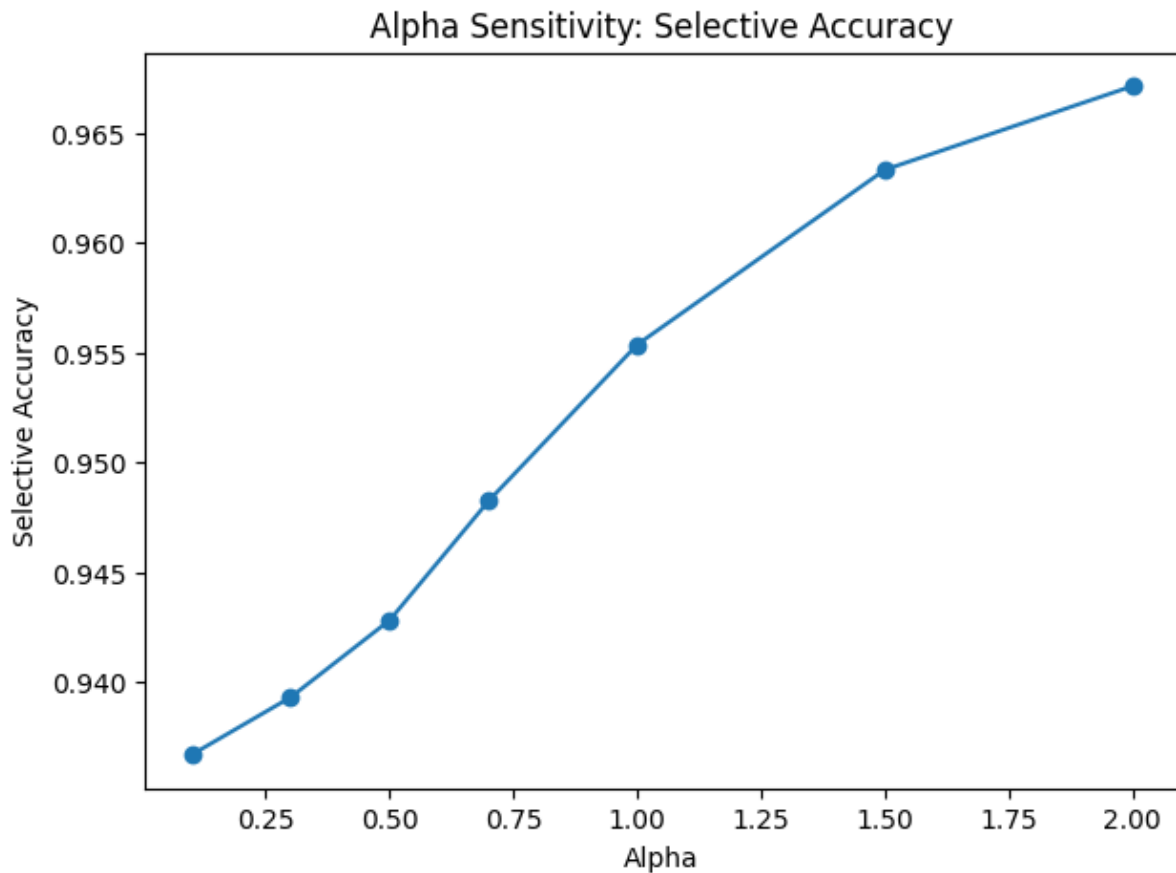


Figure 4. Effect of drift sensitivity parameter α on selective accuracy.

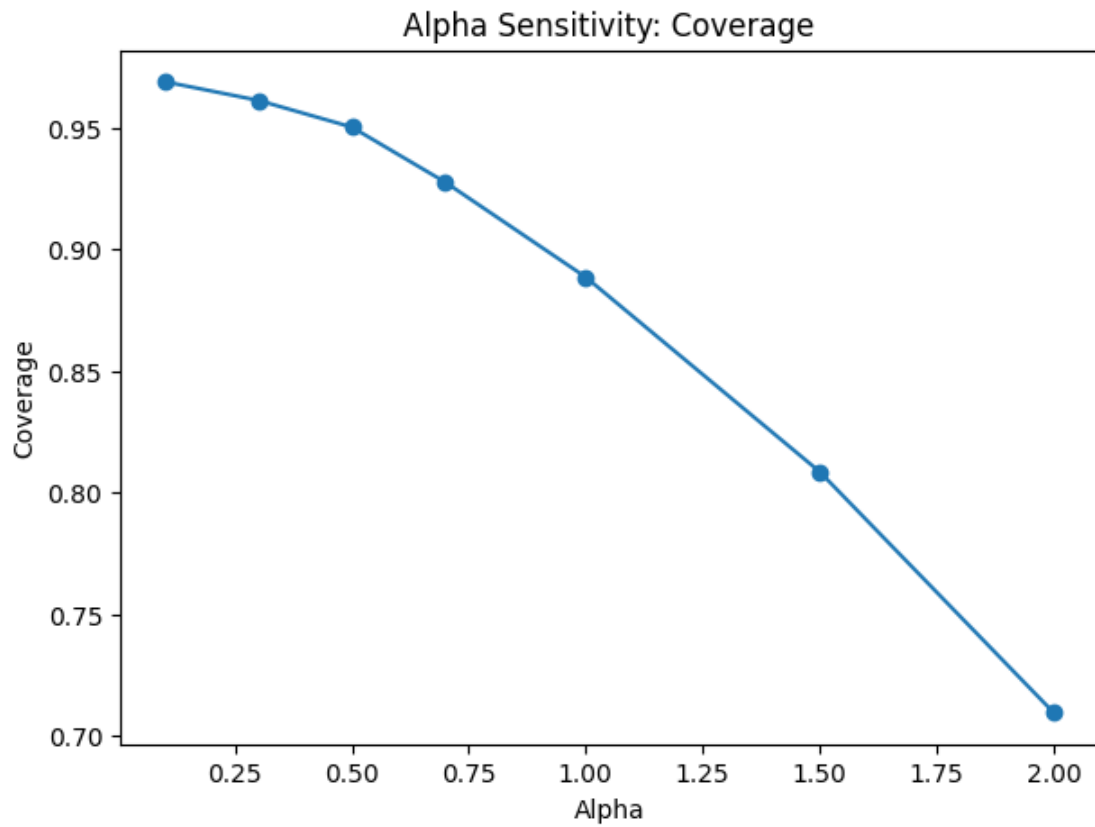


Figure 5. Effect of drift sensitivity parameter α on predictive coverage.

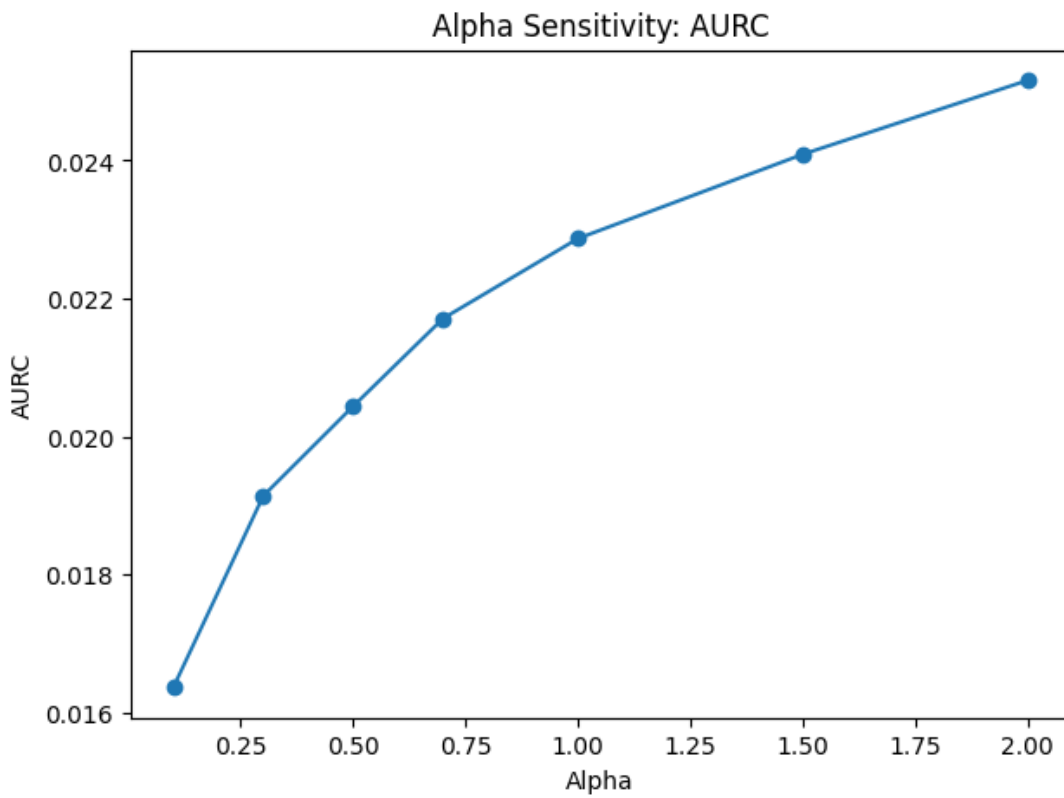


Figure 6. Effect of drift sensitivity parameter α on Area Under the Risk–Coverage Curve (AURC).

5.7 Out-of-Domain Semantic Shift Evaluation

To evaluate reliability under semantic distribution shift, the model was tested on the Yahoo Answers dataset as an out-of-domain benchmark.

The normalized OOD drift mean increased substantially to:

$$\bar{D}_{\text{OOD}} = 0.457,$$

indicating strong semantic deviation from the AG News training distribution.

Despite this substantial drift escalation, predictive confidence remained unexpectedly high:

$$\bar{c}_{\text{OOD}} = 0.911.$$

Figure 7 shows that confidence distributions remain sharply concentrated near high-confidence regions even under out-of-domain conditions.

This result provides strong empirical evidence that transformer confidence remains overconfident under semantic drift.

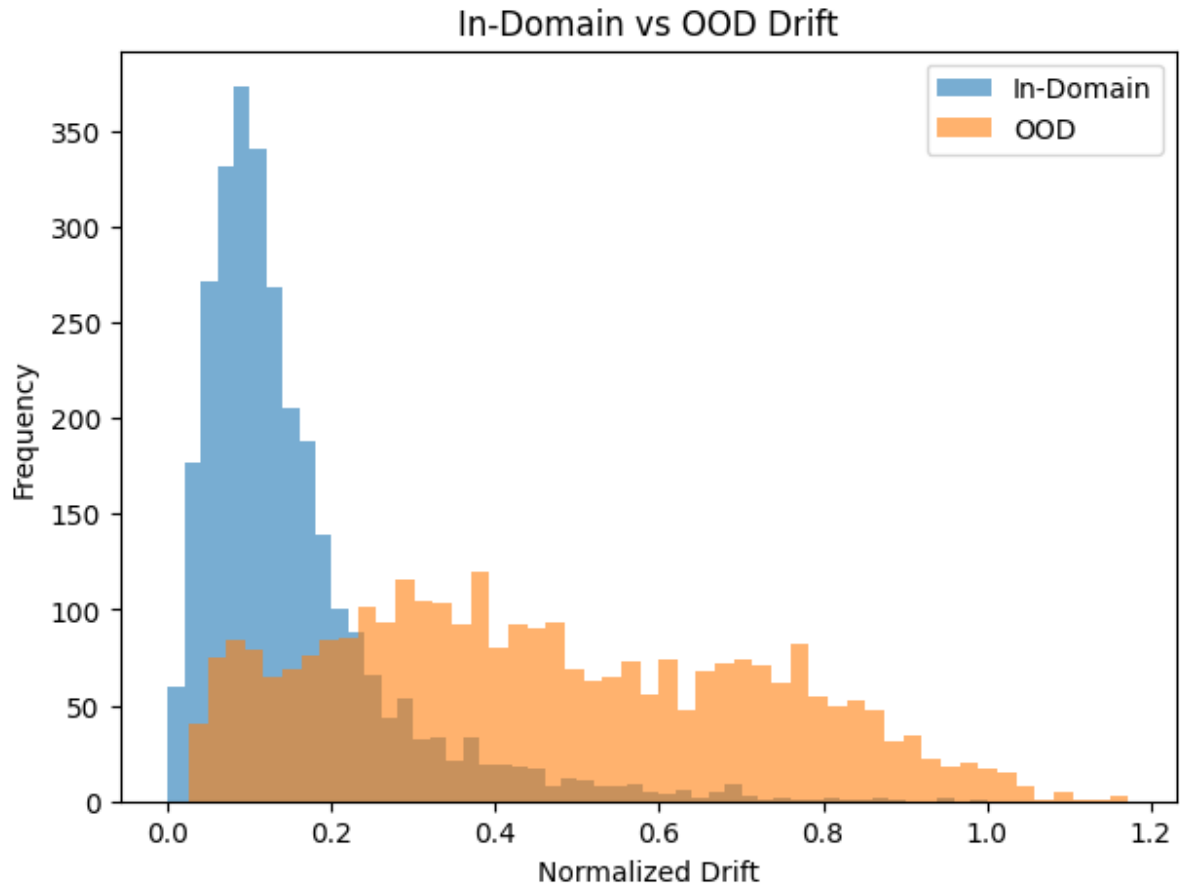


Figure 7. Comparison of normalized semantic drift distributions between in-domain and out-of-domain samples.

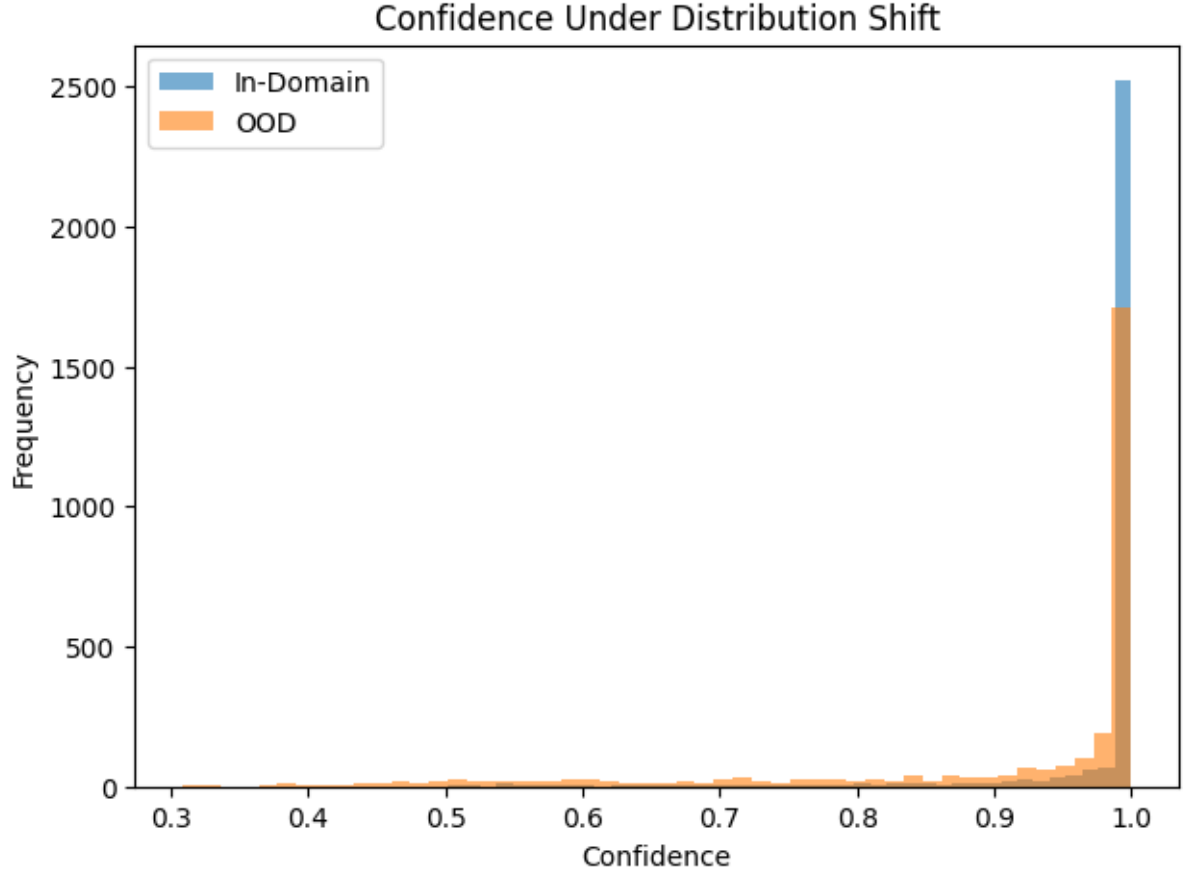


Figure 8. Predictive confidence distributions under in-domain and out-of-domain conditions.

5.8 Drift-Aware Abstention Under Distribution Shift

Finally, we evaluated the operational behaviour of DASC under out-of-domain conditions.

Using:

$$\alpha = 0.5,$$

the proposed framework reduced in-domain coverage from:

$$95.0\% \rightarrow 67.5\%$$

under semantic distribution shift.

This behaviour demonstrates that DASC selectively reduces operational trust for semantically shifted samples without requiring retraining or explicit OOD supervision.

Figure 9 further illustrates that drift-aware confidence distributions shift substantially downward under out-of-domain conditions, enabling adaptive abstention under semantic drift.

Overall, the results demonstrate that integrating semantic drift estimation with selective prediction provides an effective and lightweight reliability mechanism for transformer models under distribution shift.

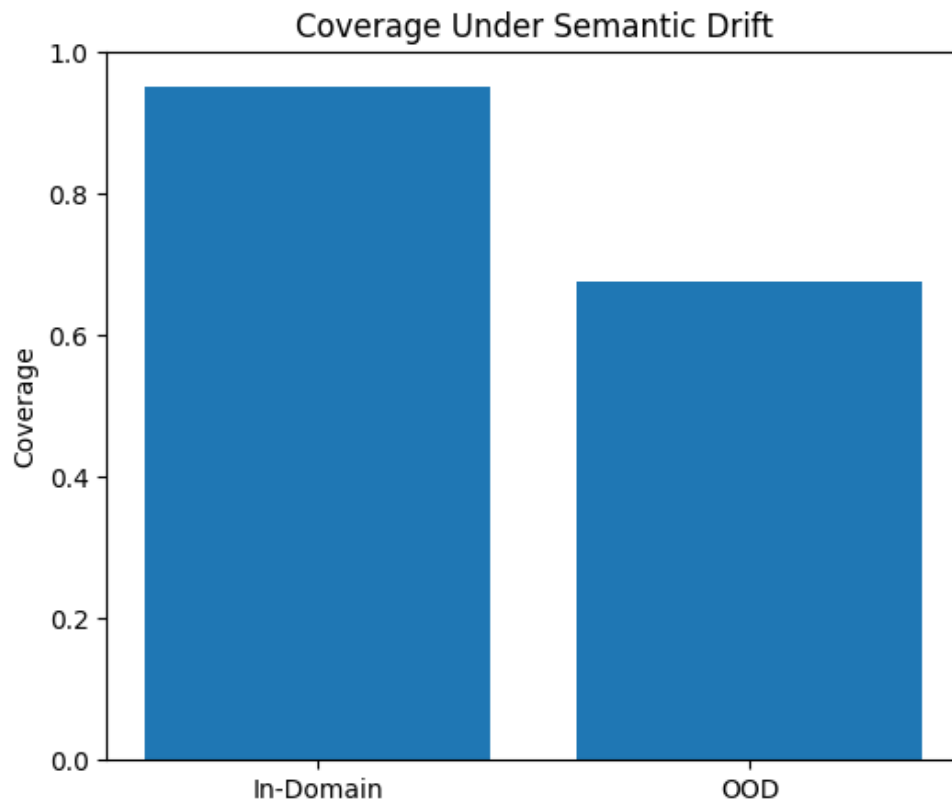


Figure 9. Predictive coverage reduction under semantic distribution shift using DASC.

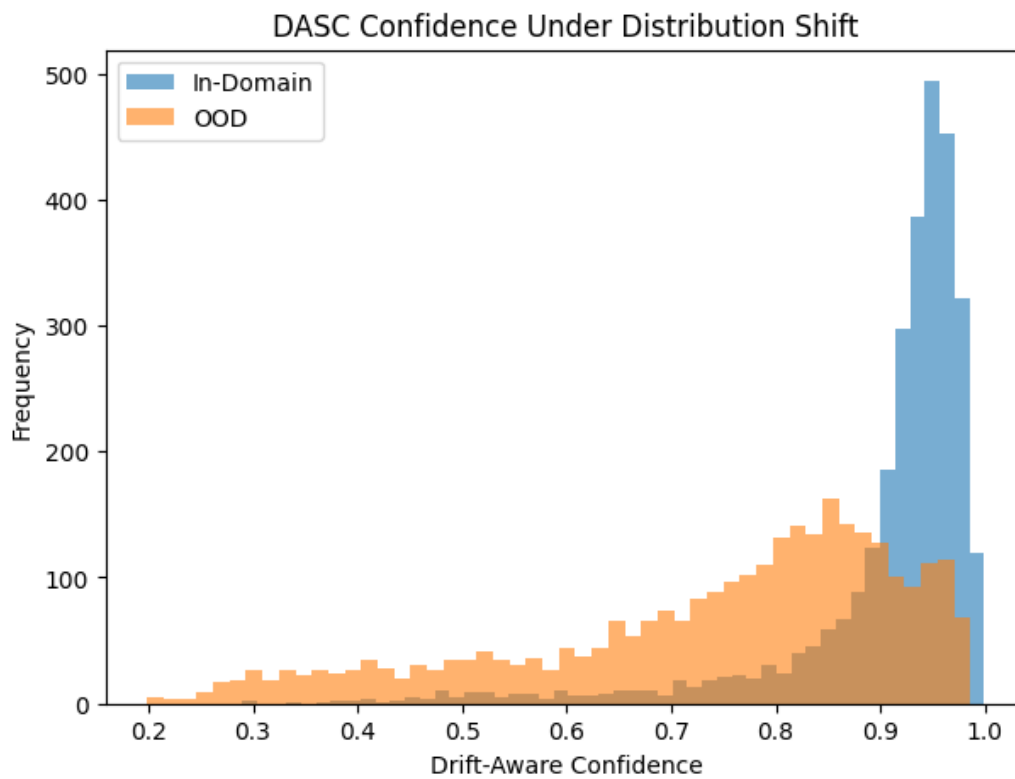


Figure 10. Drift-aware confidence distributions under in-domain and out-of-domain conditions.

6 Discussion

This work investigated transformer reliability under semantic distribution shift through the perspective of drift-aware selective prediction. The experimental results demonstrate that transformer confidence remains substantially overconfident even when representation-space semantic deviation increases significantly. This finding highlights an important limitation of conventional confidence-based reliability estimation in modern transformer architectures.

A central observation of this study is that predictive confidence alone does not reliably reflect semantic uncertainty under distribution shift. Although semantic drift increased considerably for out-of-domain Yahoo Answers samples, predictive confidence remained strongly concentrated near high-confidence regions. The observed confidence saturation under semantic shift suggests that softmax probabilities may fail to provide meaningful operational reliability signals in semantically unfamiliar environments.

The proposed Drift-Aware Selective Calibration (DASC) framework addresses this limitation by integrating representation-space semantic drift directly into selective prediction. Instead of relying exclusively on predictive confidence, the proposed method reduces operational trust as embedding-space deviation increases. Experimental results demonstrate that this approach successfully induces adaptive abstention under semantic shift without requiring retraining, architectural modification, or additional supervision.

The selective prediction experiments further revealed a controllable reliability–coverage tradeoff governed by drift sensitivity. Increasing the drift sensitivity parameter α consistently improved selective accuracy by filtering semantically unreliable predictions. However, larger α values also reduced predictive coverage and increased abstention frequency. This behaviour indicates that excessive drift penalization may induce overly conservative prediction behaviour, reducing selective efficiency despite improved retained-sample reliability.

Importantly, the results suggest that moderate drift sensitivity provides the most balanced operational behaviour. Lower α values achieved improved Area Under the Risk–Coverage Curve (AURC), while larger values maximized retained prediction accuracy. These findings indicate that drift-aware selective prediction should be interpreted as a controllable reliability mechanism rather than a universally aggressive rejection strategy.

Another important contribution of this work lies in its computational simplicity. The proposed framework operates entirely at inference time using transformer embeddings that are already produced during forward propagation. Unlike many uncertainty estimation approaches that require ensembles, Bayesian inference, adversarial training, or retraining-based calibration, DASC introduces minimal computational overhead while remaining architecture-independent. This makes the approach potentially attractive for practical deployment settings where lightweight reliability estimation is required.

The representation-space perspective adopted in this work also provides a broader interpretation of semantic reliability. Mahalanobis drift estimation captures semantic deviation at the embedding level rather than relying solely on probability distributions.

Consequently, the proposed approach can identify semantically shifted inputs even when predictive confidence remains artificially high. This relationship between embedding-space geometry and operational reliability may provide useful insights for future trustworthy-AI systems.

Despite these promising results, several limitations remain. First, experiments were conducted using a single transformer architecture and limited dataset combinations. Although the AG News–Yahoo Answers setup provides meaningful semantic shift, broader evaluation across additional domains, languages, and transformer families would strengthen generalization claims. Second, the proposed framework currently employs a relatively simple exponential drift modulation function. Alternative reliability modulation strategies, including adaptive or learned drift weighting mechanisms, may further improve selective efficiency. Third, the present work focuses primarily on text classification tasks. Extending drift-aware selective prediction to generative language models, retrieval systems, or multimodal architectures remains an important future direction.

Future research may also investigate deeper theoretical connections between semantic drift, representation-space geometry, calibration error, and prediction instability. In particular, integrating margin-based uncertainty estimation with selective abstention could provide stronger theoretical guarantees for operational reliability under distribution shift.

Overall, the findings demonstrate that semantic drift estimation provides valuable reliability information beyond predictive confidence alone. By combining embedding-space drift analysis with selective prediction, the proposed DASC framework offers a lightweight and effective mechanism for improving transformer reliability under semantic distribution shift.

Although larger drift sensitivity values increase AURC due to aggressive abstention, they simultaneously improve retained-sample reliability under semantic shift. This indicates that DASC prioritizes operational trustworthiness over maximal predictive coverage.

7 Limitations

Despite the promising results obtained in this study, several limitations should be acknowledged.

First, the experimental evaluation was conducted using a single transformer architecture, namely DistilBERT. Although DistilBERT provides an efficient and widely adopted benchmark for text classification, reliability behaviour may vary across larger transformer models and alternative architectures such as RoBERTa, DeBERTa, ELECTRA, or instruction-tuned language models. Consequently, the generalizability of the proposed framework across broader transformer families remains an open question.

Second, the current experiments were performed using a limited number of datasets. While the AG News and Yahoo Answers datasets provide meaningful semantic distribution shift, additional evaluation across diverse domains, multilingual corpora, noisy real-world datasets, and domain adaptation benchmarks would provide a more comprehensive assessment of robustness under varying semantic conditions.

Third, the proposed framework employs Mahalanobis distance as the primary semantic drift estimator. Although Mahalanobis distance effectively captures covariance-aware embedding deviation, its performance depends on the quality and stability of the estimated covariance structure. In high-dimensional embedding spaces, covariance estimation may become sensitive to dataset size and representation variability. Alternative representation-space metrics, manifold-based drift estimation methods, or learned uncertainty representations may provide improved robustness.

Fourth, the drift-aware confidence modulation function used in this work is intentionally lightweight and heuristic:

$$S(x) = c(x)\exp(-\alpha\tilde{D}(x)).$$

Although this formulation demonstrated effective reliability behaviour, the exponential modulation mechanism is not theoretically optimal and may not generalize uniformly across all tasks or datasets. More sophisticated adaptive weighting strategies or learnable drift-aware calibration mechanisms could potentially improve selective efficiency.

Fifth, the present work focuses primarily on selective prediction in text classification settings. Modern transformer systems increasingly operate in generative, retrieval-augmented, multimodal, and instruction-following environments where uncertainty behaviour may differ substantially. Extending drift-aware selective prediction to large language models and generative AI systems remains an important future research direction.

Another limitation concerns the absence of stronger selective prediction baselines such as SelectiveNet, conformal prediction methods, Bayesian uncertainty estimation, or ensemble-based selective calibration approaches. Although the objective of this work was to develop a lightweight inference-time framework, broader baseline comparison would strengthen comparative evaluation.

Finally, the current study primarily investigates empirical operational reliability rather than providing formal theoretical guarantees. While the observed relationship between semantic drift and selective risk is experimentally supported, future work could develop stronger mathematical formulations connecting representation-space geometry, calibration error, margin instability, and abstention behaviour.

Future research directions may therefore include:

- evaluation across larger transformer architectures,
- adaptive or learnable drift-aware calibration mechanisms,
- integration with conformal or Bayesian uncertainty estimation,
- multimodal and generative AI reliability analysis,
- and deeper theoretical investigation of semantic drift and selective risk relationships.

Despite these limitations, the proposed framework demonstrates that representation-space semantic drift can serve as an effective reliability signal for selective transformer prediction under distribution shift.

8 Conclusion

This work presented Drift-Aware Selective Calibration (DASC), a lightweight inference-time framework for improving transformer reliability under semantic distribution shift. Unlike conventional confidence-based selective prediction approaches, the proposed method integrates representation-space semantic drift estimation directly into the prediction process using Mahalanobis embedding-space analysis and adaptive confidence modulation.

Experimental results demonstrated that transformer models remain substantially overconfident under semantic drift, even when embedding-space deviation increases significantly. In out-of-domain evaluation using Yahoo Answers, predictive confidence remained highly concentrated near confident regions despite strong semantic shift from the AG News training distribution. These findings highlight an important limitation of confidence-only reliability estimation in transformer architectures.

The proposed DASC framework addressed this limitation by selectively reducing operational trust for semantically drifted samples. Results showed that drift-aware selective prediction improves retained-sample reliability while inducing adaptive abstention behaviour under distribution shift. Sensitivity analysis further revealed a controllable reliability–coverage tradeoff governed by drift sensitivity, demonstrating that selective reliability can be modulated according to deployment requirements.

Importantly, the proposed framework operates entirely at inference time without requiring retraining, additional supervision, ensemble methods, or architectural modification. This lightweight design makes DASC computationally practical for real-world transformer deployment scenarios where operational reliability is critical.

Overall, the findings suggest that representation-space semantic drift provides valuable reliability information beyond predictive confidence alone. By integrating semantic drift estimation with selective prediction, the proposed approach offers an effective and interpretable mechanism for improving transformer reliability under distribution shift.

Future work may extend drift-aware selective prediction toward:

- larger transformer architectures,
- multimodal and generative AI systems,
- adaptive uncertainty modulation,
- and theoretically grounded reliability guarantees under semantic drift.

The results of this study contribute toward the broader objective of developing trustworthy and uncertainty-aware transformer systems capable of safer deployment in dynamically shifting real-world environments.

9 Declarations

Competing Interests

The author declares no competing interests.

Funding Information

This research received no external funding.

Data Availability Statement

All datasets used in this study are publicly available through the HuggingFace Datasets repository, including:

- AG News
- Yahoo Answers Topics

The implementation code and experimental notebooks used to generate the reported results are available from the corresponding author upon reasonable request.

Research Involving Human and or Animals

This study does not involve human participants or animals.

Informed Consent

Not applicable.

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